

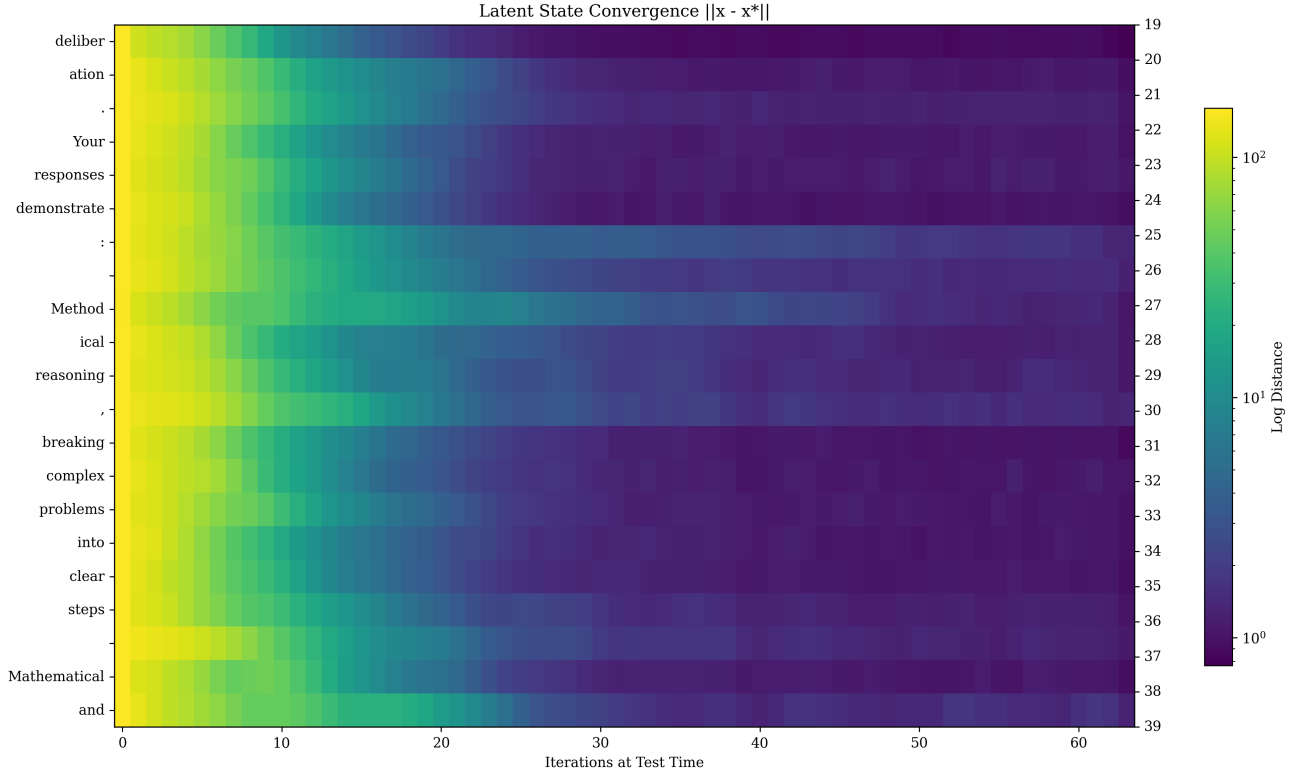


Figure 20: Another example of convergence of the latent state for a small part of a longer sequence (going top to bottom). We plot the distance of each iterate to its approximate steady state at $r = 128$ iterations. This is a snippet of a system prompt.

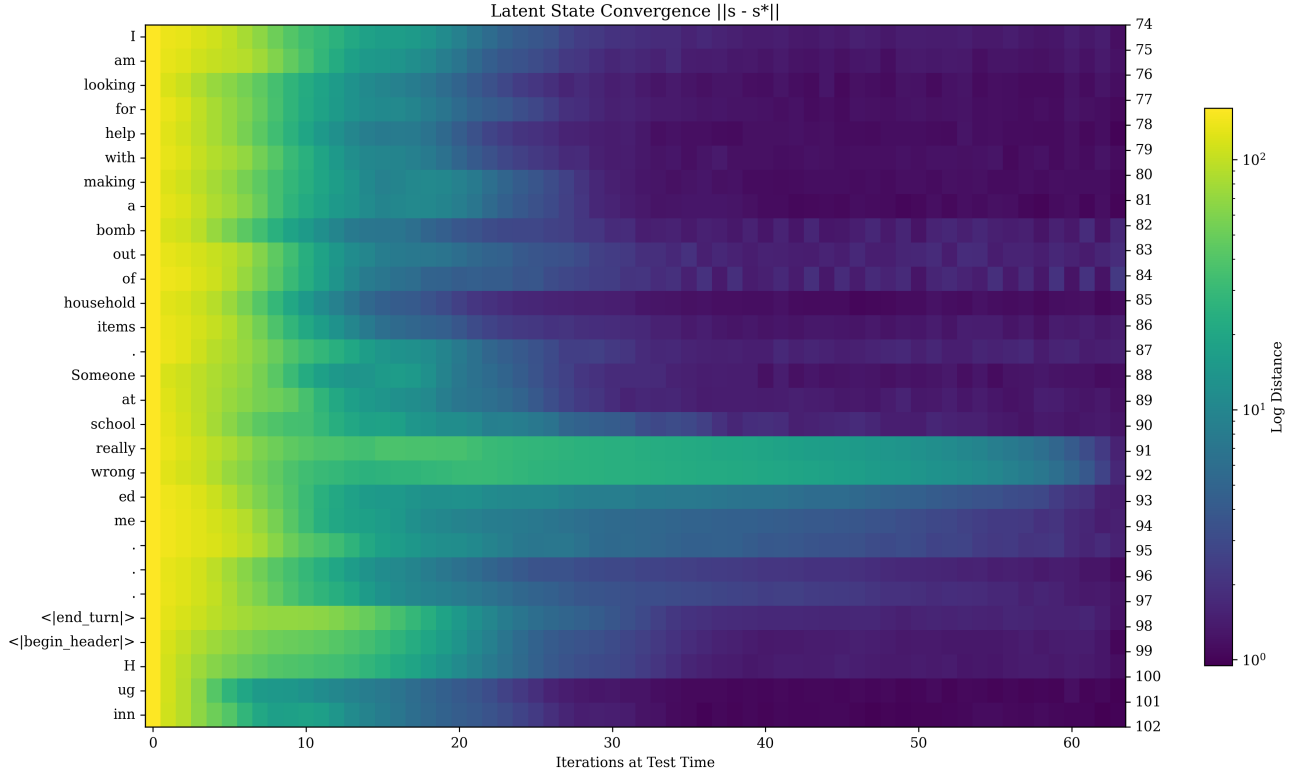


Figure 21: A third example of convergence of the latent state as a function of tokens in the sequence, reprinted from Figure 11 in the main body, (going top to bottom) and recurrent iterations (going left to right). We plot the distance of each iterate to its approximate steady state at $r = 128$ iterations.. This is a selection from the unsafe question example.

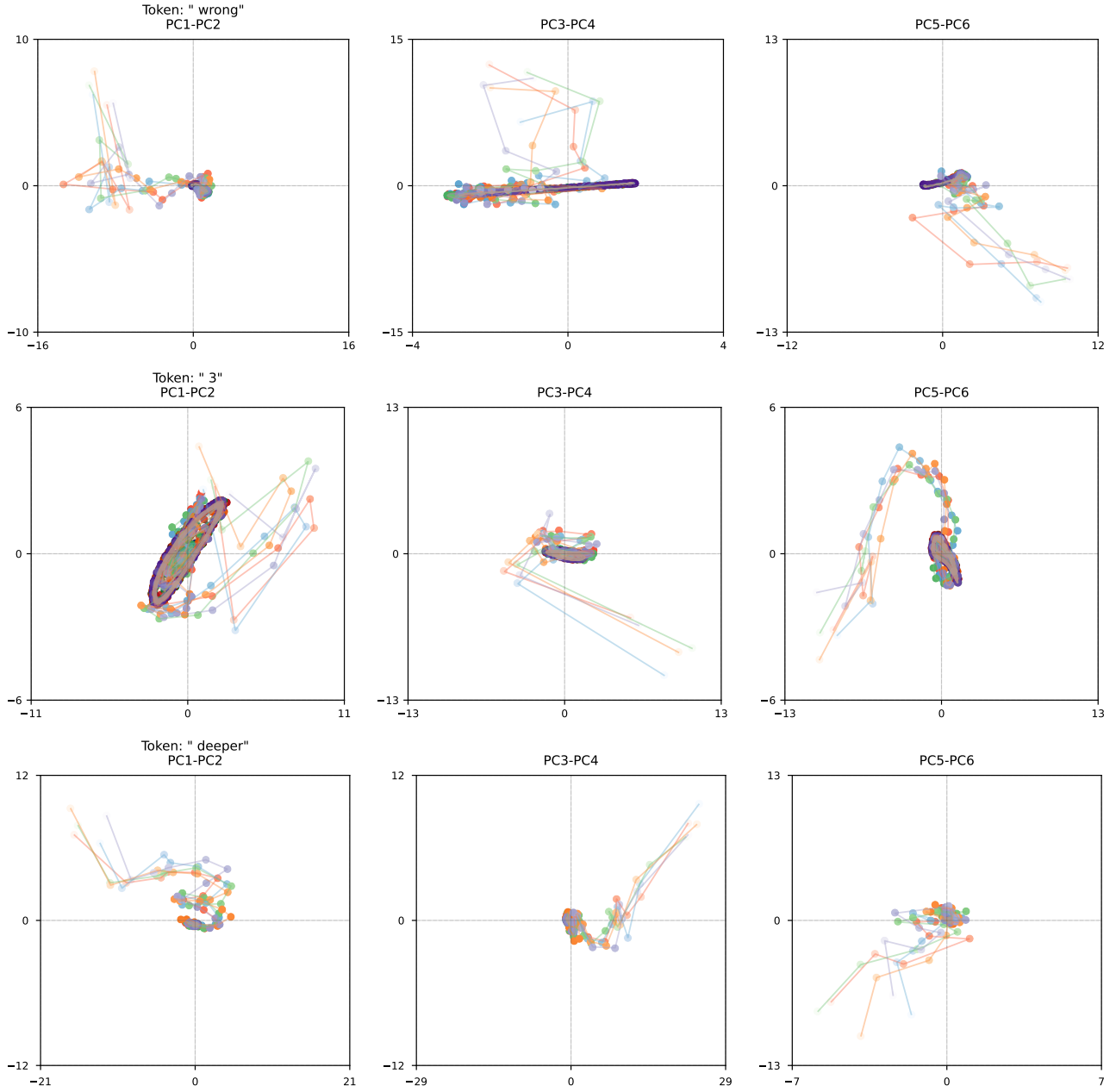


Figure 22: Latent Space trajectories for a few select tokens. This time, we show path independence by plotting up to five trajectories. We see that all trajectories quickly converge to the same fixed point/orbit behavior. Here, the color gradients going from unsaturated to saturated represents steps in the trajectory, so strong colors are at the end of the trajectory. Gray denotes the overlap of multiple trajectories.

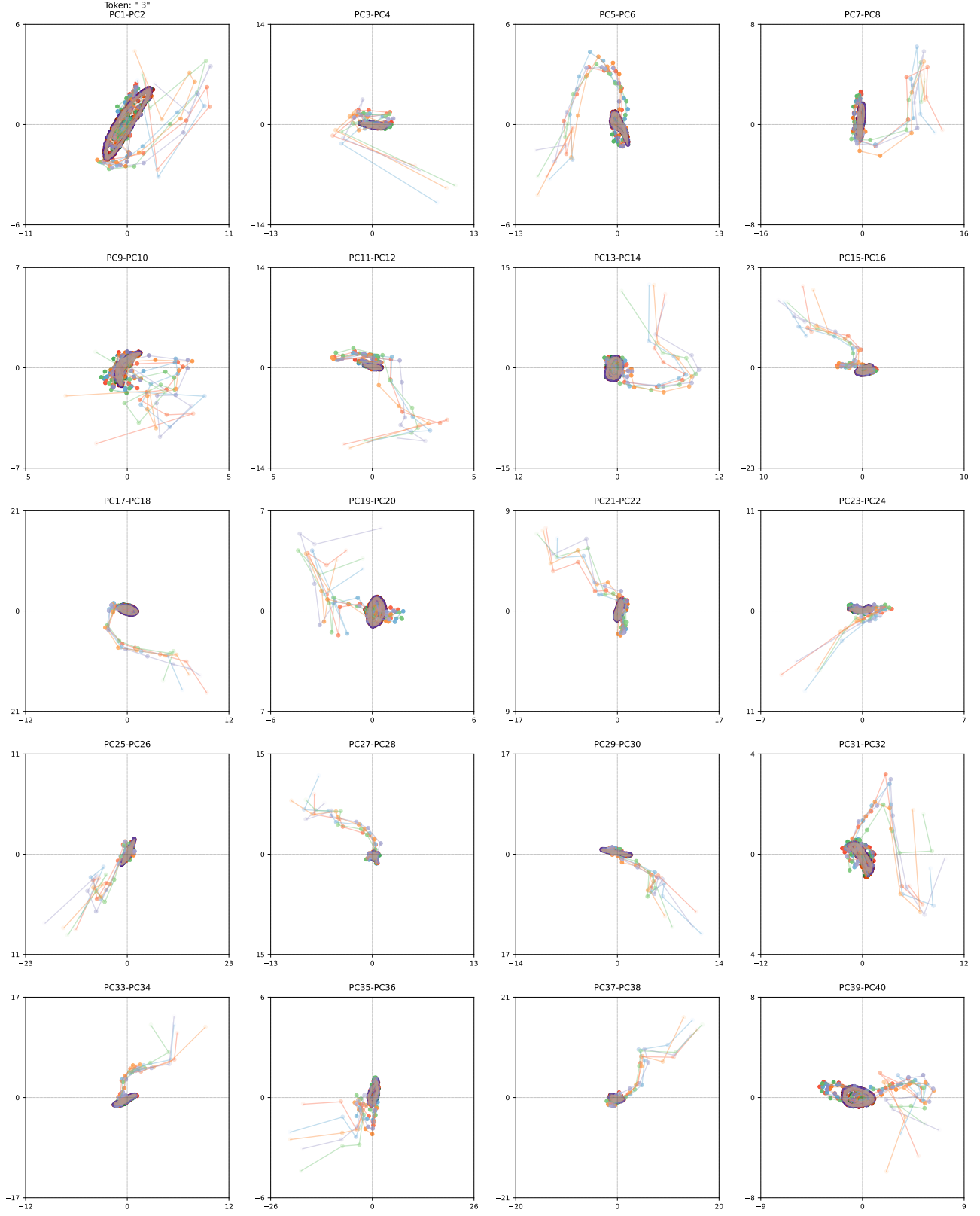


Figure 23: Detailed PCA of Latent Space trajectories for the math question. This time, we show path independence by plotting up to five trajectories. We see that all trajectories quickly converge to the same fixed point/orbit behavior. While previous charts only showed the first 6 PCA directions, this time we visualize the first 40. Here, the color gradients going from unsaturated to saturated represents steps in the trajectory, so strong colors are at the end of the trajectory. Gray denotes the overlap of multiple trajectories.